Homework #4

Question 1.

a)
```
boolean checkBST (Node root) {
        if (root.left() == null && root.right() == null)
                return true;
        if (root.left() != null) {
                if (root.right() != null)
                        return root.left().value() < root.value() && root.right().value() >
                                root.value() && checkBST(root.left()) &&
                                checkBST(root.right());
                return root.left().value() < root.value() && checkBST(root.left()) &&
                                checkBST(root.right());
                }
        return root.right().value() < root.value() && checkBST(root.left()) &&
                checkBST(root.right());
}
```

b)
```
int height (Node root) {
        if (root == null)
                return -1;

int leftHeight = height(root.left());
int rightHeight = height(root.right());

return Math.max(leftHeight, rightHeight) + 1;
}

boolean isAVL (Node root) {
        return checkBST(root) && AVLHelper(root);
}

boolean AVLHelper (Node root) {
        if (root == null)
                return true;

        int difference = height(root.left()) - (height(root.right));

        if (difference < -1 || difference > 1)
                return false;

        return AVLHelper(root.left()) && AVLHelper(root.right());
```

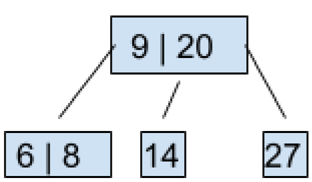
```
c)     T                          V                           V
      / \          →             / \            →            / \
     K   Y                      T   W                       K   W
                               / \   \                     / \   \
                              K   U   Y                   G   T   Y
                                                         /   / \
                                                        E   N   U
```

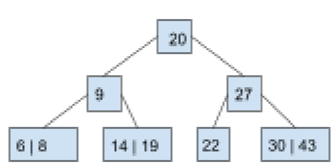
```
d)          U
           / \
          T   V
         /     \
        K       W
       / \       \
      G   N       Y
     /
    E
```

```
e)      E
         \
          T
         / \
        G   U
         \   \
          K   V
           \   \
            N   W
                 \
                  Y
```

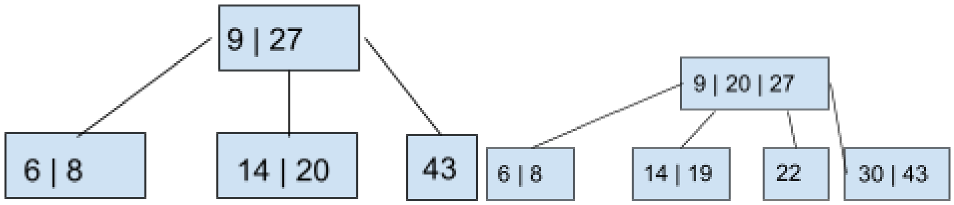
Question 2.

a)

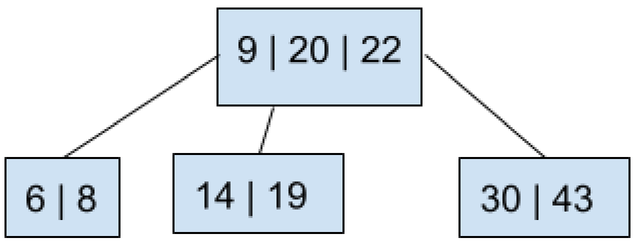




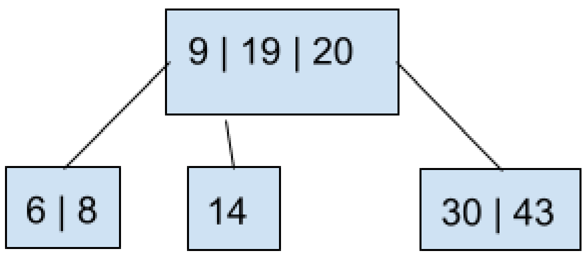
b)


9 | 27
6 | 8
14 | 20
43
9 | 20 | 27
6 | 8
14 | 19
22
30 | 43

c)


9 | 20 | 22
6 | 8
14 | 19
30 | 43

d)


9 | 19 | 20
6 | 8
14
30 | 43